J Evol Econ (2011) 21:373–406
DOI 10.1007/s00191-011-0224-6

REGULAR ARTICLE

Financial and economic determinants of firm default

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Published online: 19 February 2011
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Abstract This paper investigates the relevance of financial and economic variables as determinants of firm default. Our analysis covers a large sample of medium-sized limited liability firms. Since default might lead, through bankruptcy or radical restructuring, to firm's exit, our work also relates to previous contributions on industrial demography. Using non parametric tests we assess to what extent defaulting firms differ from the non-defaulting group. Bootstrap probit regressions confirm that economic variables, in addition to standard financial indicators, play both a long and short term effect. Our findings are robust with respect to the inclusion of Distance to Default and risk ratings among the regressors.

Keywords Firm default · Selection and growth dynamics · Stochastic equality · Bootstrap probit regressions · Distance to Default · Credit ratings

JEL Classification C14 · C25 · D20 · G30 · L11

We would like to thank the attendants of the "Revolving Doors" conference for useful suggestions on a preliminary version of the paper. The final version benefits from the insightful hints of Elena Cefis and two anonymous referees. The research leading to these results has received funding from the European Community's Seventh Framework Programme (FP7/2007-2013) under Socio-economic Sciences and Humanities, grant agreement n° 217466.

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1 Introduction

Business failures have been the subject of extensive analysis in applied economics and econometrics. One obvious reason is that the "death" of a firm represents the ultimate market response to the inability of an economic activity to survive the competitive pressure. In this sense the study of business failures essentially rests on the more general issue of identifying the characteristics which make a given firm, or a group of firms, competitive. At the same time, the failure of a business activity represents a costly waste of tangible and intangible assets, something that entrepreneurs, managers, firms' creditors, employees and the society as a whole would surely prefer to avoid. Clearly, our ability to design policies in order to minimize the occurrence, or the effects, of this unwanted event depends on our understanding of the original and contingent determinants of firm death.

Traditionally, industrial economic literature studied firm's death focusing on exit events as reported in business registers. From an economic point of view, however, these events are often spurious. They might be in fact associated with a simple relabeling of the economic subject, following a change of ownership or a modification of incorporation status. These changes do not necessarily imply any economic effect on business operations, nor signal any actual difficulty on the part of the firm. Moreover, even when the exit is real, it can take various forms, like voluntary liquidation, acquisition or bankruptcy. These forms of exit represent very different economic outcomes and are likely to be caused by different factors. Despite these differences, most of the existing literature has treated exit as a homogeneous event (see, for instance, Dunne et al. 1988, Mata and Portugal 1994 and Disney et al. 2003). Recent contributions (Honjo 2000; Cefis and Marsili 2007; Esteve-Pèrez et al. 2010) try and distinguish between the different routes to exit, but they are still limited by the difficult interpretation of the related legal events (restructuring, acquisition, merger, etc.) and by their exact timing. Indeed, when a firm is about to exit the market, information gets scarcer, and it becomes difficult to determine with precision when operations actually ceased.[1] Of course being unable to exactly determine the time of the exit severely constraints the possibility to realize large scale studies (a case to the point are, for instance, the works by Schary 1991; Agarwal and Gort 1996; Audretsch et al. 1999).

In this paper we use a different strategy to study the issue. We exploit information on distress events occurring in a large panel of Italian firms and identify potential business failures with financial defaults. A default is considered to occur when the obligor is past due more than 90 days or when the lending bank considers that the obligor is unlikely to repay its debt in full

[1] Bankruptcy procedures might procrastinate for years before they conclude and the exiting firm is eventually deleted from the register. In Italy, for instance, it took on average more than 7 years, in 2004, for bankruptcy procedures to get to an end (ISTAT 2006).

(BIS 2006).[2] Like firm failures, default events are both a signal of business troubles and a costly condition that should be in principle avoided. Even if default is not directly related with exit, it constitutes the main requisite for initiating the bankruptcy procedures.[3] If a defaulting firm is not filed for bankruptcy[4] it is then very likely that a process of profound restructuring takes place, often resulting into a substantial change in the ownership of the firm (Cefis and Marsili 2007; Hotckiss et al. 2008), or that the firm is interested by some form of merger and acquisition (M&A). As a result, the event of default is strictly related to the "involuntary" modes of exit identified in the literature of industrial demography (Schary 1991). Moreover, since firms' solvency conditions are strictly monitored by lending banks, the default event is precisely identified in time.

The firm default has been extensively studied in the literature devoted to banking and corporate finance, where it is used as one of the main signals of financial distress. This literature has traditionally investigated the relation of default events with various financial indicators (leverage, cost of debt, debt time structure, etc.), often with the aim at developing a synthetic rating measure capable to account for the risk involved in financing a particular business activity (see Beaver 1966 and Altman 1968; and Altman and Saunders 1998 and Crouhy et al. 2000 for reviews). In these models firm distress is typically conceived to be primarily determined by poor financial conditions, especially in the short run before default occurs. Purely industrial factors tend to receive less attention and sometimes turn completely left out of the analysis. The best example is provided by the development of the "Distance to Default" (DD) model which prices the market value of a firm as a call option on the value of the firm, with strike price given by the face value of its liabilities (Black and Scholes 1973; Merton 1974). The idea is that the owner exercises the right to operate the firm if the (properly discounted) expected future revenues generated by the activities are higher enough to offset the liabilities and the revenues obtained with the present liquidation of the assets. Under the assumption of efficient markets, the value of firm's equity, the market values of its assets and the face value of its debt are enough to derive the probability to default over a given time horizon.

In the present work we start by focusing on those variables that are generally proposed in the literature of financial economics as explanatory of firms

[2]Such conditions might slightly vary, depending on the internal regulations of every single bank. Since we consider here only default events collected by a single bank, we confront ourselves with a homogeneous definition.

[3]According to the Italian civil law there is a subjective and an objective requisite for accessing bankruptcy law. The former is the professional nature of the activity and the latter is insolvency. The subjects entitled to appeal for the application of bankruptcy law are the firm itself, the bank or other creditors.

[4]The evidence suggests that the subjects involved do not always find it convenient to apply for bankruptcy law (Shrieves and Stevens 1979; Gilson et al. 1990; Gilson 1997; Hotckiss et al. 2008).

default. Yet, while the financial nature of default events clearly suggests to primarily look for short-term financial causes, it is well understood that the probability to stay in the market, as well as the financial stability of a firm, is deeply intertwined with the ability to perform well along the economic or industrial dimensions of its operation. Thus, it is likely that looking exclusively at financial indicators cannot offer but a partial account of the main determinants of default, at least as long as market frictions or other institutional factors are affecting the extent to and the speed at which economic performances get reflected into financial structure and financial conditions.[5] For this reason we will augment our initial purely financial models with a set of economic predictors, including proxies for productivity, profitability, size and growth, which are likely to be significantly related with firms' success (and failure). Indeed, despite different schools of thought exist on the theory of the firm, there is strong agreement that the variables above represent the key levels of corporate performance along the process of competition and selection.[6]

A remarkable feature of the present analysis rests on its broad scope. While typical studies of firm default focus on large publicly traded companies, our study covers almost 20,000 limited liabilities firms active in manufacturing, very different in size and type of activity. This makes our study highly representative of the dynamics of the Italian industry. Moreover, the simoultaneous consideration of financial and economic variables allows us to bridge the financial and industrial literature on, respectively, default and exit.[7]

We offer two specific contributions. First, we explore the heterogeneities possibly existing both within and across defaulting vis á vis non-defaulting firms along each dimension considered. We compare the empirical distribution of financial and economic characteristics across the two groups, and employ non-parametric tests for stochastic equality to assess distributional differences, both close to the default event and in the years before it occurs. Second, and in accordance with the main purpose of the paper, we estimate a series of probit models of default probability, allowing us to identify the interplay of economic and financial factors as determinants of default. Bootstrap techniques are used to obtain robust estimates of the relevant coefficients, and a set of model evaluation criteria is also introduced, enabling to discern if default prediction accuracy is improved when economic characteristics are added to financial factors.

[5]Starting from similar considerations Grunert et al. (2005) propose an augmented version of a standard financial model of default prediction which also includes two "soft" non-financial characteristics (managerial quality and market position) among the regressors.

[6]The same message is consistent with broad sense neoclassical models of firm-industry dynamics (see, for instance, Jovanovic 1982; Ericson and Pakes 1995; Melitz 2003), as well with models originating from the evolutionary tradition (see Winter 1971; Nelson and Winter 1982).

[7]A huge empirical literature has highlighted the positive effect exerted on survival by the technological characteristics of the firms, like R&D expenditures or patents (see Agarwal and Audretsch 2001, for a review). Unfortunately we lack the necessary data to include these further dimensions in our analysis (see details in Section 2).

The analysis of empirical distributions reveals that defaulting and non-defaulting firms display important differences along both economic and financial characteristics. Further, results from bootstrap probit regressions reveal that the impact on the probability of default exerted by economic characteristics remains significant even near to default, when one would instead expect that economic factors have been already embodied into financial conditions. The findings do not change when we add, among the regressors, an approximate measure of Distance to Default and an official credit rating index. This supports the robustness of our conclusions with respect to the inclusion of further dimensions which we might not directly capture through the set of financial and economic variables available to us. All the results do not depend from sectoral specificity at the level of 2-Digit industries.

The remaining of the article is organized as follows. Section 2 presents a detailed description of the dataset. A first descriptive comparison of defaulting *vs* non-defaulting firms is provided in Section 3, based on kernel estimates of the empirical distributions of economic and financial variables in the two groups of firms. Section 4 further explores the issue by means of more formal statistical tests of distributional equality. In Section 5 we tackle bootstrap probit estimates of default probability, focusing on whether the addition of economic variables can improve the explanatory and predictive power of the models, as compared to a benchmark specification where only financial indicators are used. Robustness of results with respect to the inclusion of Distance to Default and credit ratings is then tested in Section 6. Finally, Section 7 concludes suggesting some interpretations.

2 Data, variables and sample selection

The present analysis is based on a list of default events occuring within Italian manufacturing in either 2003 or 2004. These events are independently provided by an Italian bank (labeled "Bank" in what follows) only for those firms which were among its customers. The list is linked with information obtained from the Italian Company Account Data Service (Centrale dei Bilanci, CeBi) database, which contains financial statements and balance sheets of virtually all Italian *limited liability* firms. Italian Civil Law enforces the public availability of the annual accounting for this category of firms. CeBi collects and organizes this information, performing initial reliability checks. The database is quite rich and detailed and includes firms operating in all industrial sectors without any threshold imposed on their size. This represents a remarkable advantage over other firm level panels, which typically cover only firms reporting more than a certain number of employees, and makes it particularly suitable for the analysis of both large and small-medium sized firms.

Merging the list of defaults made available by the Bank with CeBi data provides the final dataset. This contains balance sheet information for manufacturing firms over the period 1998-2003 with the addition of a dummy variable taking on value 1 when a firm appears in our list, i.e. incurs default

at the end of the period (in either 2003 or 2004), and 0 otherwise.[8] Since the list contains only customers of the Bank, some end-of-period defaulters can be present in the CeBi database, which are not in our list. It is therefore likely that our dataset understates the frequency rates of default actually occurring in the population. This is a common situation in studies of distress prediction, potentially problematic for regression analysis, as incorrect estimates might arise due to a classical "choice-based sample bias".[9] A priori, one would like to be sure that defaulting customers of the Bank are representative of the entire population of Italian limited liability defaulting firms, meaning that there are no systematic differences between observed and non-observed defaulters. Since we do not have access to information on other defaulters we have to resort to indirect evidence. First consider that, in most developed countries, it is nowadays common for firms to establish multiple banking relationships, that is debt relationships with more than one bank (see Ongena and Smith 2000). The available evidence tells that this tendency is particularly strong in Italy, where Foglia et al. (1998) and Cosci and Meliciani (2002) report that the median number of lending banks per firm, computed in the same CeBi dataset we are using, ranges between 6 and 8. This suggests a very low probability that the particular features that we identify in the analysis only apply to the subset of customers of our Bank. Moreover, and following Bank's specific advise, we limited the scope of analysis to a subset of CeBi firms which are more comparable to the Bank's customers. This restricts the sample to only include those firms reporting at least 1 Million Euro of Total Sales in each year and more than one employee.[10]

The resulting sample includes 19,628 manufacturing firms and 147 default events. An important piece of information is that such default events amount to approximately 20% of the defaults taking place in those years in the reference population of limited firms. This is not a small number, providing further indication that our defaulters are unlikely to differ from other defaulters. Table 1 shows that, however, under-weighting of distressed events is only partially solved by the implemented cuts. The default rates in our data are apparently lower than default rates in the reference population of Italian limited liability firms (averaged between 2003 and 2004), as officially reported by the association of the Italian Chambers of Commerce, by 2-Digit sectors.[11]

[8] The identity of firms has not be disclosed to us. The matching procedure was performed directly by the Bank.

[9] Zmijevski (1984) analyzes this point in depth. Notice however that default events tend to be over-represented in the samples typically employed in that literature, an opposite situation as compared to the problem we must face here.

[10] At the same time these cuts allow to focus the study only on firms displaying at least a minimal level of structure and operation.

[11] Sectors are defined according to the NACE (Rev.1.1) industrial classification *Nomenclature génerale des Activités économiques dans les Communautés Europénnes*, which is the standard at European level, and perfectly matches, at the 2-Digit level, with the International Standard Industrial Classification, ISIC.

Table 1 Sectoral default rates

Sectors	Number of firms	Number of defaults	Default rate in the sample	UNIONCAMERE default rate
15—Food products & beverages	2,008	9	0.0045	0.0302
17—Manufacture of textiles	1,544	14	0.0091	0.0474
18—Wearing apparel; dressing; dyeing of fur	673	9	0.0134	0.0511
19—Leather; luggage, & footwear	762	15	0.0197	0.0375
20—Manufacture of wood	396	2	0.0051	0.0249
21—Pulp & paper products	540	2	0.0037	0.0298
22—Publishing, printing & recorded media	749	6	0.0080	0.0377
24—Chemical products	1,122	2	0.0018	0.0383
25—Rubber and plastic products	1,176	6	0.0051	0.0338
26—Other non-metallic mineral products	1,142	5	0.0044	0.0309
27—Manufacture of basic metals	672	2	0.0030	0.0378
28—Metal products (except machinery & equip.)	2,473	14	0.0057	0.0280
29—Machinery and equipment n.e.c.	2,916	23	0.0079	0.0352
31—Electrical machinery and apparatus n.e.c.	835	7	0.0084	0.0361
32—Radio, TV & communication equip.	280	6	0.0214	0.0473
33—Medical, precision & optical instruments	472	1	0.0021	0.0399
34—Motor vehicles, trailers & semi-trailers	442	8	0.0181	0.0364
35—Other transport equipment	234	6	0.0256	0.0382
36—Furniture; manufacturing n.e.c.	1,184	10	0.0084	0.0341
Total	19,628	147	0.0075	

The first three columns report respectively the number of firms, the number of defaults default rates in the sample, computed at 2-Digit sectoral level. The last column displays the corresponding default rates in the population of Italian limited liability firms (averages between 2003 and 2004. Source: The Association of the Italian Chambers of Commerce, UNIONCAMERE)

Since this problem is likely to be particularly harmful for probit estimates, the analyses in Sections 5 and 6 also apply a bootstrap sampling procedure designed to make default frequencies equivalent to the actual default rates observed at the population level.

We use a set of financial and economic variables available through CeBi balance sheets data. On the financial side, although we can build less indicators than one typically finds in studies of bankruptcy prediction, we can anyway capture the "strands of intuition" (see Carey and Hrycay 2001) lying behind this type of analysis by focusing on three variables: Interest Expenses (IE), which provides a flow measure of the annual costs borne by firms to repay debt; Leverage (measured as the ratio of Total Assets over the sum of the value of outstanding shares plus income after taxes), which is a standard indicator of the relative balance between external vs. internal financing; and, finally, the Financial-Debt-to-Sales ratio (FD/S), which gives a stock measure of overall exposure, scaled by the size of the firm. To cure the potential problems arising from the limited number of financial variables we will perform robustness checks augmenting our models with an official rating index provided by CeBi. Rating procedures are indeed designed to embrace a wider range of firms' characteristics, together with qualitative and quantitative assessment of industry as well as national scenarios, technological changes, regulatory framework, and so on.[12] The CeBi index is an issuer credit rating, meaning that it gives an assessment of the obligor's overall capacity to meet its obligations, without implying any specific judgment about the quality of a particular liability of the company. It is updated at the end of each year, and thus allowed to change over time. The method employed for the computation of the index is exclusive property of CeBi. There is, however, no reason to expect that the procedure is dramatically different from the methods applied by other rating agencies, both in terms of being targeted over the very short run (as said, one year ahead) and in terms of embracing a wide range of firms' characteristics. Finally as a further measure of firm overall financial soundness, we will consider a proxy of Distance to Default (DD). Distance to Default is at the core of the last generation of empirical models of default prediction, adopted both by scholars and practitioners.[13] The theoretical foundation of this measure derives from an application of classical finance theory (Black and Scholes 1973; Merton 1974), modeling the market value of firm equity as a call option on the value of the firm, with strike price given by the face value of its liabilities. Accordingly, DD

[12]This is typically the case with credit ratings issued by international agencies (see the "prototype risk rating system" described in Crouhy et al. 2001). This tendency has been more recently confirmed by the effect of the provisions of the Basel II process, encouraging banks and financial institutions to introduce ratings-based internal systems of risk assessment which consider a broad and multidimensional evaluation of their exposure (see BIS 2001).

[13]See Duffie et al. (2007), for recent advances in financial literature, and the works cited therein for a review of duration models based on Distance to Default. Crosbie and Bohn (2003) offer an extensive introduction to Moody's KMV model, which is also based on Distance to Default theory.

is defined as a function of firms' underlying value of assets, of the volatility of the latter and of the face value of debt. Under standard assumptions, the probability of default is obtained as the value of a standardized Normal density computed in DD, which is therefore a sufficient statistic to predict default.

Concerning economic characteristics, we are able to include in the analysis measures of size, growth, profitability and productivity, that is a basic set of dimensions which theoretical models as well as empirical research in industrial economics suggests to capture the crucial indicators of firm performances. Several proxies can be in principle adopted to measure each of these dimensions, each proxy capturing complementary aspects of the same phenomenon. First, concerning firm size, revenue based (sales or value added) measures appear to be more suited to have a relationship with default than alternative "physical" (in terms of employment or capital) measures. Thus, we measure size in terms of Total Sales (S), and, accordingly, the growth rate of Total Sales (g^S) is used to measure firm growth. Second, concerning profitability, we want a proxy of the margins generated by the industrial or operational activities of the firms, which is the level we are interested into, avoiding definitions of income or profits influenced by financial strategies and taxation. Accordingly, we define profitability in terms of the return on sales (ROS), i.e. the ratio between Gross Operating Income and Total Sales. Finally, productive efficiency is captured by a standard index of labor productivity, measured in terms of value added per employee. Using labor productivity and gross operating income we are focusing on the more operative part of firm activity. We have no access to the kind of data typically used to estimate the stock of productive capital (like tangible assets). Consider however that while the cost and age of capital endowments certainly play a role in defining firm productive capabilities, their cumulative effect is already reflected in the financial variables we are considering.[14]

3 Descriptive analysis

This section analyzes if and to what extent the economic and financial characteristics of defaulting firms differ from the rest of the sample. We compare the empirical distribution of the relevant variables across the two groups of firms. Notice that the estimates of kernel densities provide information on properties of the variables, such as shape, degrees of heterogeneity among different classes of firms, skewness, etc., which are usually ignored by financial studies specialized in default prediction. To take account of the possible intertemporal variation, we present results at different time distance to default, comparing densities in the first available year, 1998, with densities obtained in the last year before default occurs, 2002. We apply non-parametric techniques, which

[14]The short horizon of our analysis implies that capital intensity of each firm is basically fixed over the sample period.

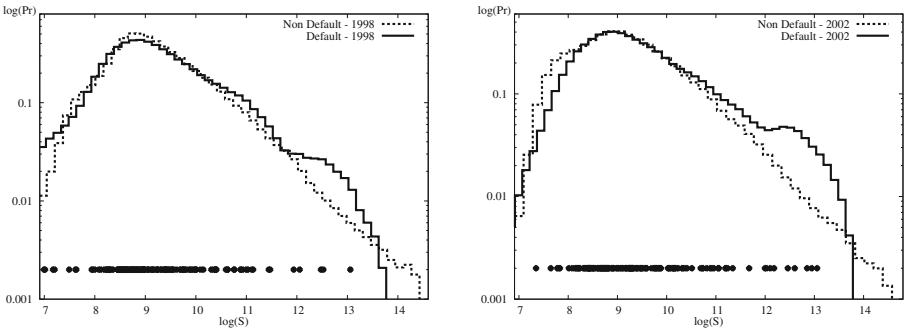


Fig. 1 Empirical density of total sales (S) in 1998 (*left*) and 2002 (*right*): defaulting vs. non-defaulting firms

do not impose any a priori structure to the data, thereby allowing to take a fresh look at the heterogeneities possibly existing both within and across the two groups of firms. The descriptive nature of this analysis is supplemented by formal statistical tests for distributional equality, performed in the next section.

3.1 Economic characteristics

We start with the comparison of firm size. In the two panels of Fig. 1 we plot, on a double logarithmic scale, the kernel density of firm size (S) estimated for defaulting and non-defaulting firms in 1998 and in 2002. The actual values of S for each defaulting firm are depicted in the bottom part of each plot.[15]

First look at 1998. Somewhat contrary to what one might conjecture, defaulting firms are neither less heterogeneous nor smaller with respect to the rest of the sample. The two densities are indeed very similar: supports are comparable, and the shapes are both right-skewed, an empirical result repeatedly found in the literature on firm size distribution. Actually, the right part of defaulting firms' density suggests that default events are more frequent among medium-big sized firms rather than at small sizes. Estimates for 2002, in the right panel, show that these properties remain stable over time, as the default is approaching. The upper tail of the defaulting firms' distribution is indeed even heavier as compared to 1998. A formal non-parametric test of multi-modality (Silverman 1986) cannot reject the presence of bimodality in the distributions of defaulting firms (with a p-score of 0.72 for 1998 and of 0.63 for 2002). Such differences in the tail behavior are anyhow due to relatively few very big firms (see the dots at the bottom of the plots). Instead, in the central part of the densities, where most of the observations are placed, the overlap is almost perfect. Overall, the evidence is therefore suggestive that there is no

[15]Here, as well as in the following, estimates are performed applying an Epanenchnikov kernel, and the bandwidth is set following the "optimal rules" suggested in Silverman (1986, Section 3.4) .

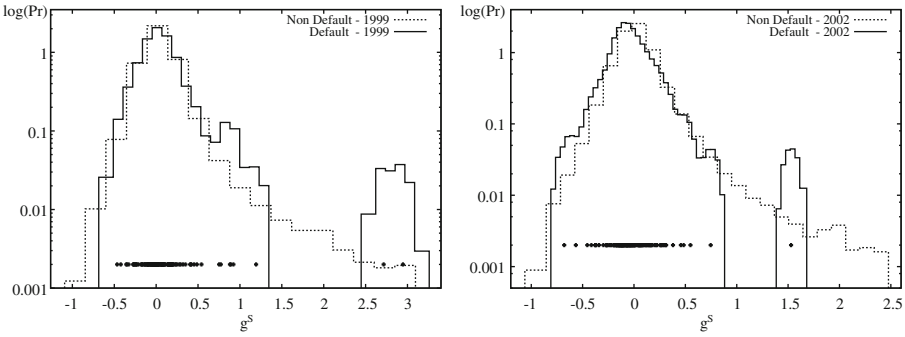


Fig. 2 Empirical density of Total Sales Growth (g^S) in 1999 (*left*) and 2002 (*right*): defaulting vs. non-defaulting firms

clearcut relationship between size and the event of default: operating above a certain size threshold does not seem to provide any relevant warranty in preventing default.

Next, Fig. 2 compares kernel densities of Total Sales growth rates, g^S. Given the initial year in the sample is 1998, the first available data point is for 1999. This is shown in the left panel, while 2002 is depicted on the right graph. Actual values of g^S for defaulting firms are reported below the estimated densities.

Defaulting firms do not appear to significantly differ from the rest of the sample when considering the portions of supports where most of the probability mass is concentrated (approximately between -0.5 and 0.5). In this interval, the distributions are crossing each other, and the estimated shapes are very similar, independently from the different time distance to default considered. One difference emerges regarding the variability of growth episodes: in 2002, that is closer to the default event, the width of the supports spanned by the defaulting group is sensibly narrower, especially in the right part of the distributions. This gets also mirrored in the tails (outside the interval $[-0.5, 0.5]$), where we however observe some differences across the two groups. In both years considered, left tail behavior is similar across the two groups, suggesting similar occurrence of extremely bad growth records. Conversely, only few defaulting firms are responsible for the peaks present at the top extremes. The different sizes of the two compared samples is likely to play a role in this respect. Similarly to what noted for size, however, these tail patterns concern a very low number of firms, and therefore they offer too weak evidence to conclude that one of the two groups is significantly outperforming the other.

We then repeat the same exercise with profitability performance. Figure 3 reports kernel estimates of ROS densities in 1998 and 2002. The two groups of firms tend this time to differ, as defaulting firms perform clearly worse than the rest of the sample, especially for positive values of ROS. In 1998, the two distributions are substantially overlapping in the negative half of the

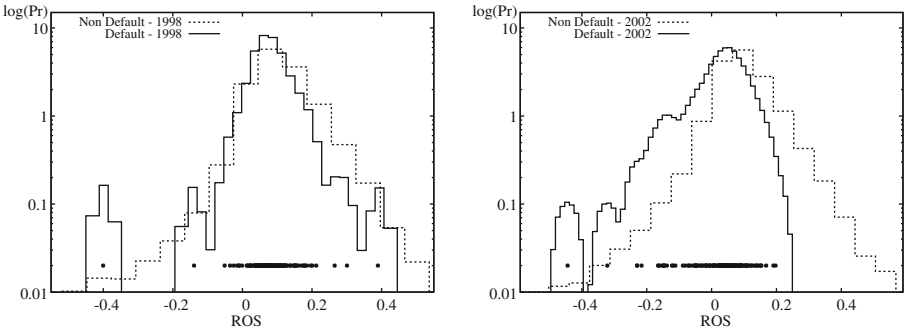


Fig. 3 Empirical density of Profitability (ROS) in 1998 (*left*) and 2002 (*right*): defaulting vs. non-defaulting firms

support, while the density of defaulting firms lies constantly below that of the other group in the positive half. The same ranking gets reinforced in 2002. The distance between the two distributions in the right part of the support increases, and the density of defaulting firms is much concentrated at negative values. Despite negative performance is experienced also by non-defaulting firms, the evidence suggests that a sort of selection on profitability is at work: default events tend to be associated with lower profitability levels. In addition, time plays an important role in the story, since profitability differentials across the two groups tend to become wider in the very short run before default.

Finally, the densities of Labour Productivity, plotted in Fig. 4, show that a similar mechanism is also acting upon productive efficiency. Indeed, the estimates obtained for non-defaulting firms tend to lie to the right of the ones obtained for defaulting firms, especially if one nets out the effect of few outliers present at the extremes. The intertemporal patterns also resembles the findings observed for profitability: the productivity advantage of non-defaulting firms increases over time. This suggests that Labour Productivity

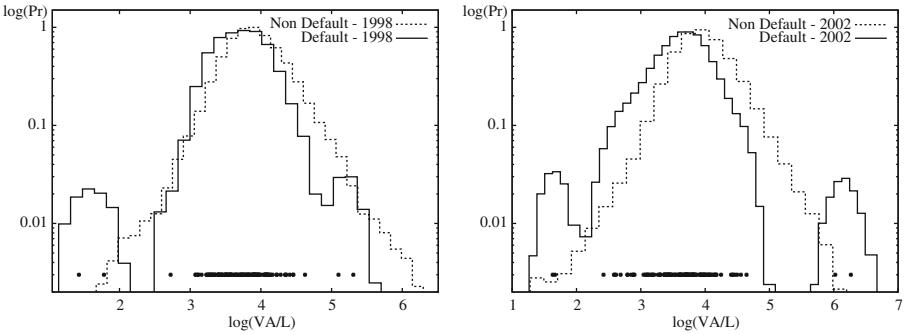


Fig. 4 Empirical density of Labour Productivity (VA/L) in 1998 (*left*) and 2002 (*right*): defaulting vs. non-defaulting firms

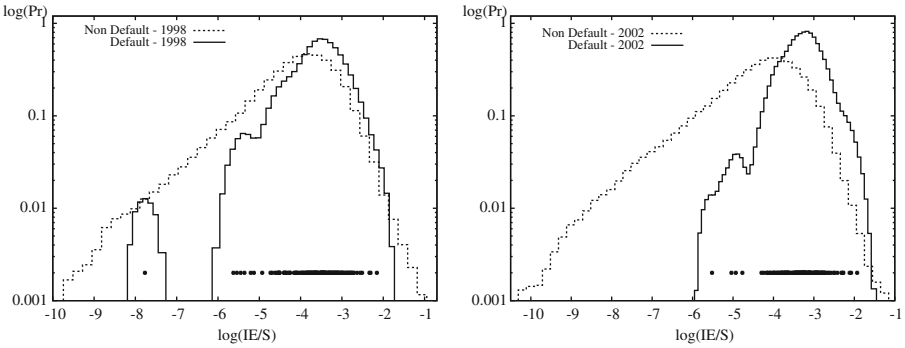


Fig. 5 Empirical density of Interest Expenses scaled by size (IE/S) in 1998 (*left*) and 2002 (*right*): defaulting vs. non-defaulting firms

represents a discriminatory factor telling apart defaulting firms from the rest of the sample. The relevance of this factor seems increasing as the default event approaches.

3.2 Financial characteristics

We then ask whether defaulting firms display any significant peculiarity in terms of financial indicators. Figure 5 shows densities of Interest Expenses over Sales (IE/S), i.e. the proportion of annual revenues that goes to meet interest payments. The resulting estimates, reported in logs, suggest a clearcut difference between the two groups, as indeed defaulting firms display larger average and modal values. Also, the shape of the distributions differ, with the defaulting firms much more concentrated in the right part of the support. A further noticeable feature is that, whereas the estimates for non-defaulting firms do not change over time, the density of defaulting firms displays a rightward shift of probability mass between 1998 and 2002. This means that the

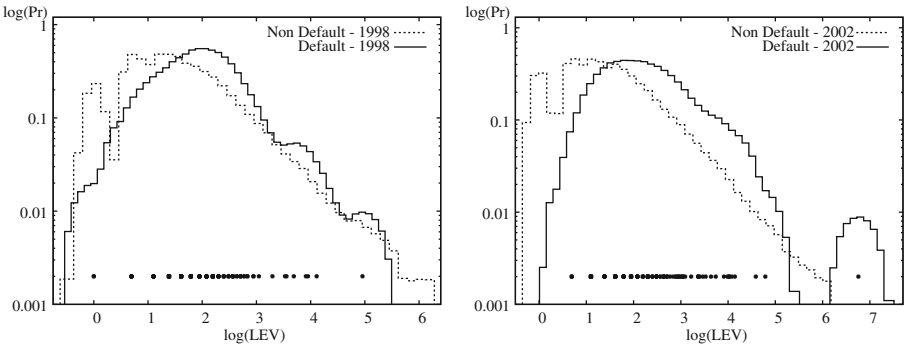


Fig. 6 Empirical density of Leverage (LEV) in 1998 (*left*) and 2002 (*right*): defaulting vs. non-defaulting firms

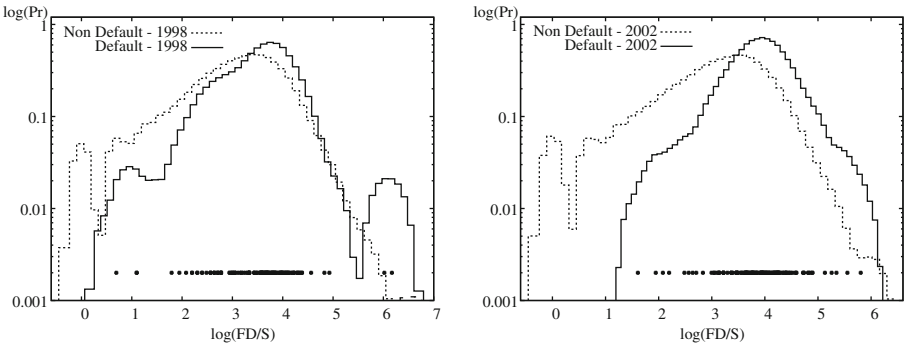


Fig. 7 Empirical density of Debt-to-Sales ratio (FD/S) in 1998 (*left*) and 2002 (*right*): defaulting vs. non-defaulting firms

flows of interest payments per unit of output sold becomes heavier as the default event approaches.

The densities of Leverage (Fig. 6) and Financial Debt-to-Sales ratio (Fig. 7) follow similar inter-temporal dynamics. The rightward shift in the Leverage distribution of defaulting firms indicates that the ratio between external vs. own resources increases over time, resulting into a disproportionate financial structure in proximity of the default event. At the same time, the even more remarkable shift in the FD/S ratio complement the above results on IE/S: not only the flow of debt repayment, but also the stock of debt is increasing when default approaches. Also notice that the differences between defaulting and non-defaulting firms, in terms of both Leverage and FD/S ratios, are smaller than in terms of IE/S. This possibly signals that cost of debt is the financial factor which more sharply distinguish defaulters from non-defaulters.

4 Non-parametric inferential analysis

In order to add statistical precision to the comparison between the two groups of firms, we now perform formal tests of distributional equality. A range of testing procedures is in principle available. There are however some specific features of our data which must be carefully considered in selecting the most appropriate alternative. First of all, default events are much less frequent than non-defaults, and therefore we need a test which can be applied in the case of two uneven samples. Second, as shown in the previous section, the distributions we are going to compare display clear non-normalities and unequal variances, suggesting that non-parametric tests should be preferred over parametric ones. Further, even within the class of non-parametric tests for comparison of uneven samples, a common feature is to implicitly assume that the samples to be compared only differ for a shift of location, while their distributions possess identical shapes. Given that equality of shapes is generally violated by our data, as shown by kernel densities, it is appropriate to employ

tests which abandon this hypothesis. When distributions with different shapes are compared, looking at the relative location of medians, modes or means might no longer be very informative, as the very meaning of these measures changes with the nature of the underlying distribution. In this case a better measure of the relative position of two samples is provided by the idea of stochastic (in)equality.

Let F_D and F_{ND} be the distributions of a given economic or financial variable, for the two samples respectively. Denote with $\mathbf{X}_D \sim F_D$ and $\mathbf{X}_{ND} \sim F_{ND}$ the associated random variables, and with X_D and X_{ND} two respective realizations. The distribution F_D is said to dominate F_{ND} if $\text{Prob}\{X_D > X_{ND}\} > 1/2$. That is, if one randomly selects two firms, one from the D group and one from the ND group, the probability that the latter displays a smaller value of X is more than $1/2$, or, in other terms, it has a higher probability to have the smaller value. Now, since

$$\text{Prob}\{X_D > X_{ND}\} = \int dF_D(X)\, F_{ND}(X), \tag{1}$$

a statistical procedure to assess which of the two distributions dominates can be formulated as a test of

$$H_0 : \int dF_D\, F_{ND} = \frac{1}{2} \quad \text{vs.} \quad H_1 : \int dF_D\, F_{ND} \neq \frac{1}{2}. \tag{2}$$

The quantity $\hat{U}$ proposed in Fligner and Policello (1981) provides a valid statistic for H_0. We apply their procedure exploiting the fact that, in case of rejection of the null, the sign of the Fligner-Policello (FP) statistic tells which of the two groups is dominant: a positive sign means that defaulting firms have a higher probability to take on larger values of a given financial or economic variable, and the opposite holds for a negative sign of the statistic.[16]

Table 2 presents the results obtained year by year. The high rate of rejection of H_0 supports the evidence provided by the previous descriptive analysis, confirming that the two groups differ under many respects. First, looking at financial variables, the signs of the FP statistics are consistent with the idea that defaulting firms present weaker performances than non-defaulters, under all the dimensions considered. Second, as far as economic variables are concerned, defaulting firms tend to be less profitable and less productive than those in the other group, and we tend to confirm that, possibly due to the already mentioned tail behavior observed in the empirical distribution of Total Sales, defaulting firms are comparatively bigger. Growth rates instead display statistically significant differences in 2002 only.

Overall, the findings broadly confirm the conclusions based on the kernel estimates. Notice also that differences in both economic and financial performances matter over both the shorter and the longer run. With the exception of

[16]Under the further assumption that the two compared distributions are symmetric, testing H_0 is equivalent to testing for equality of medians between possibly heteroskedastic samples. This is what is usually referred to as the Fligner–Policello test (see Bottazzi et al. 2008, for details).

Table 2 Fligner–Policello test of stochastic equality, defaulting vs. non-defaulting firms

Variable	Test	Test of stochastic equality				
		1998	1999	2000	2001	2002
IE/S	FP stat	**7.510**	**9.269**	**13.019**	**17.903**	**24.069**
	p-value	0.000	0.000	0.000	0.000	0.000
LEV	FP stat	**8.029**	**10.483**	**12.066**	**13.520**	**15.190**
	p-value	0.000	0.000	0.000	0.000	0.000
FD/S	FP stat	**7.490**	**10.480**	**14.387**	**16.037**	**17.229**
	p-value	0.000	0.000	0.000	0.000	0.000
TS	FP stat	0.364	1.555	**3.988**	**3.466**	**2.426**
	p-value	0.716	0.120	0.000	0.000	0.015
GROWTH	FP stat		0.905	−0.618	−1.133	**−3.927**
	p-value		0.365	0.536	0.257	0.000
PROF	FP stat	**−4.609**	**−7.169**	**−7.186**	**−7.466**	**−11.176**
	p-value	0.000	0.000	0.000	0.000	0.000
PROD	FP stat	**−5.310**	**−7.156**	**−7.167**	**−6.842**	**−8.855**
	p-value	0.000	0.000	0.000	0.000	0.000

Observed value of the statistic (FP) and associated *p*-value. Rejection of the null means that the two distributions are different in probability. Rejection at 5% confidence level highlighted with bold

growth, the null of equality is already rejected at the beginning of the period, or at least some years before default.

5 Robust probit analysis of default probabilities

The analyses conducted so far tell us how defaulting firms compare with non-defaulting firms when each economic or financial dimension is considered on its own. In this section we try to identify which are the main determinants of default once the effects of economic and financial factors are allowed to simultaneously interplay.

To this end, we frame our research question so as to single out the effects of financial and economic variables within a more standard parametric setting. The response probability of observing the default event is modeled as a binary outcome Y (taking value 1 if default occurs, 0 otherwise), and then estimated conditional upon a set X of explanatory variables and controls. We employ a probit model, where the default probability is assumed to depend upon the covariates X only through a linear combination of the latter, $X\beta$, which is in turn mapped into the response probability through

$$\text{Prob}\,(Y = 1 \mid X) = \Phi(X\beta), \tag{3}$$

where $\Phi(\cdot)$ is the cumulative distribution function of a standard normal variable, with associated density $\phi(\cdot)$.

Several variations of Eq. 3 are explored in the following, including different sets of regressors. The estimation strategy is however common to all the specifications, and it is intended to solve the under-weighting of default events

in our data, as compared to distress rates observed in the reference population of Italian limited liability firms (recall Table 1). As anticipated, this feature of the dataset is dangerous for regression analysis, since it might give rise to a classical "choice-based sample" bias (Manski and McFadden 1981), a well known problem in studies of default probability since Zmijevski (1984). There exist different methods to get rid of the potential bias, either by employing specific estimators designed for this situation (see Manski and Lerman 1977; Imbens 1992; Cosslet 1993), or by performing bootstrap sampling. We follow this second alternative, which has the advantage that it does not depend on specific assumptions about the distribution of the estimated parameters. The only requirement is that each bootstrap sample needs to be representative of the population, and the way this is achieved is explained in the following. Studies of distress prediction, where oversampling of default events is the typical situation, achieve this goal by performing randomized re-sampling of both defaulting and non defaulting firms in the desired, population-wide proportions (see, for instance Grunert et al. 2005). In our case, the relatively low number of defaults available in the data suggests to take defaulting firms fixed, and randomly extract a subset of non-defaulting firms only. This is the strategy we apply in the following. In particular, in order to reduce the bias as much as possible, sampling of non-defaulting firms is implemented with replacement within each 2-Digit industry, so that the ratio of defaulting over non-defaulting firms equals the population-wide default frequency reported in Table 1 at this level of sectoral aggregation. The sampling procedure is repeated several times, and estimates of the different specifications of Eq. 3 are repeated on the sample obtained at each round. Averaging over the number of runs then yields robust estimates. We will present results based on 200 independent replications, which turned out to be a large enough bootstrap sample to achieve convergence in the estimated coefficients.

One problem remaining out of our direct control concerns the fact that, due to the way data are collected, some of the firms treated as non-defaulters could in fact be defaulting firms. Two considerations are due here. On the one hand, the possible presence of defaulting firms into our control group of non-defaulters implies that, whenever a variable has a statistically different effect between the two groups, the "true" difference would be even more significant if we could precisely identify non-defaulters. Thus, we can safely comment on our results when a variable turns significant.[17] On the other hand, it could be

[17]This is standard in controlled experiments. Consider for instance that you want to test if a given drug is effective. You treat a group of people for one month and then compare the result with an untreated group. Suppose you find significant differences, and therefore conclude that the drug is actually effective. Now if somebody in the control group had some doses of the drug, this of course testify *in favour* of the effectiveness of the drug, not against it: those control subjects who were not in contact with the drug were different enough to suggest an effective treatment. Coming back to our problem: if we find significant differences comparing the characteristics of defaulters and the control group of non-defaulters, then these differences would be *even more significant* if we could eliminate defaulters from the control group.

that variables that do not appear to have significantly different effects between the two groups, have indeed different effects, but such differences have been hidden by the presence of defaulting firms in the control group. Here is where our re-sampling scheme really helps. Indeed, remember that defaults occur with low frequency in the reference population (c.f. Table 1). Therefore, the probability to have a defaulter in the control group must be very small, and Monte Carlo methods are well known to be robust with respect to this kind of disturbance. So even the occurrence of this second problem can be considered remote.

Our main goal is to test the commonly held presumption that default is mainly determined by poor financial conditions, especially in the short run before default occurs. Thus, our choice of the specifications of Eq. 3 is primarily meant to verify whether adding economic variables, in general, and looking at their effect at different time distances to default, in particular, might improve the chance to correctly distinguish healthy firms from those at risk of default. Our conjecture is that explicit consideration of economic variables should improve the understanding of default dynamics.

Accordingly, we focus on comparing results of two main specifications. The first model includes, among the regressors, only financial indicators, together with a full set of sectoral (2-Digit) control dummies

$$\begin{aligned}&\text{Prob}\left(Y_T = 1 \mid X_t\right)\\ &= \Phi\left(\beta_{0t} + \beta_{1t}\,\frac{IE_t}{S_t} + \beta_{2t}\; LEV_t + \beta_{3t}\,\frac{FD_t}{S_t} + \delta_t\,\text{Sector}_t\right),\end{aligned} \tag{4}$$

where, as in the previous sections, IE/S stands for Interest Expenses scaled by Total Sales, LEV is Leverage, and FD/S is the Financial-Debt-to-Sales ratio. In the second specification we then add the economic variables

$$\begin{aligned}\text{Prob}\left(Y_T = 1 \mid X_t\right) = \Phi\Bigg(&\beta_{0t} + \beta_{1t}\,\frac{IE_t}{S_t} + \beta_{2t}\; LEV_t + \beta_{3t}\,\frac{FD_t}{S_t} + \beta_{4t}\;\ln S_t\\ &+ \beta_{5t}\; PROD_t + \beta_{6t}\; PROF_t + \beta_{7t}\; GROWTH_t\\ &+ \delta_t\,\text{Sector}_t\Bigg),\end{aligned} \tag{5}$$

where S is size (again in terms of Total Sales), PROD is Labour Productivity (as Value Added per employee), PROF is profitability (in terms of Return on Sales), and GROWTH is the log-difference of Total Sales. Recall that, due to the characteristics of the dataset, the covariates can be measured over the different years of the window 1999–2002, while the default event is only measured at the end of the period (at time labeled as T).[18] Thus, comparing estimates in the different years allows to capture the dynamic effects of the covariates on the probability of default at different time distances to the

[18] Recall that 1998 is excluded simply because growth rates cannot be computed for that year.

default event. This is a relevant issue, especially in understanding the extent to which financial conditions are indeed capturing the past history of economic dimensions of firm performance.

In panel A of Table 3 we show results obtained in each year, averaging over the 200 bootstrap replications. Columns 1–4 concern estimates of the model in Eq. 4 while columns 5–8 refer to the specification in Eq. 5. Notice that all models are estimated taking z-scores of the covariates. This reduces them to have equal (zero) mean and equal (unitary) variance, allowing for a direct comparison of the magnitudes of the estimated effects across different models. We report marginal effects, computed as standard at the sample mean of the covariates, which is zero given z-scoring. Statistical significance is assessed through confidence intervals based on bootstrap percentiles (see Efron and Tibshirani 1993). That is, we first estimate the empirical probability distribution function (EDF) of the coefficients obtained over the 200 bootstrap runs. Then, statistical significance at the α% level is rejected if the zero falls within an interval

$$\left[\hat{q}(\alpha/2), \hat{q}(1-\alpha/2)\right], \tag{6}$$

where $\hat{q}(\alpha)$ stands for the estimate of α-th quantile of the bootstrap distribution estimated from the EDF.[19] Estimates of sectoral dummies are not reported, as we indeed find that only less than 5% of these coefficients turns statistically significant at the 5% confidence level. Moreover, the few significant sectors tend to differ across the different exercises considered. These results yield strong support that sectoral specificities do not affect the link between the probability of default and the set of economic and financial characteristics included in our analysis.

The estimates corresponding to the "financial variable only" model in Eq. 4 show that cost of debt is the most relevant financial dimension. We indeed find that the relatively big and positive effect of IE/S is significant over the entire period, while Leverage and FD/S display weaker significance. This relates to the interesting variation over time of the estimates. The stock of debt tends indeed to be more relevant at longer distance to default, then loosing significance in the shorter run, when the estimated impact of the IE/S increases remarkably. Notice also that Leverage is turning significant only in the last year before default. This possibly captures part of the short run effect played by an excessive debt burden, thereby compensating for the disappearing significance of FD/S.

[19]Several refinements of the bootstrap estimates of confidence intervals are discussed in the literature, most notably the BC_a and ABC corrections. These methods require an estimate of the bias, which we can only obtain by performing a "first step" probit regression on the overall original sample. This is however exactly what we want to avoid, in order to overcome under-sampling of defaulting firms. Alternatively, one could try to estimate the bias by re-sampling from each random sample. This second order bootstrap seems to us unnecessary due to the relatively large size of the sample considered.

Table 3 Probit estimates of default probabilities as modeled in Eqs. 4 and 5—results over 200 bootstrap replications

	Bootstrapped probit regressions—estimates by year							
	(1) 1999	(2) 2000	(3) 2001	(4) 2002	(5) 1999	(6) 2000	(7) 2001	(8) 2002
Panel A: estimation results								
IE/S	0.0051*	0.0056*	0.0073*	0.0139*	0.0054*	0.0055*	0.0071*	0.0130*
LEV	0.0039	0.0032	0.0026	0.0063*	0.0026	0.0020	0.0014	0.0057*
FD/S	0.0067*	0.0080	0.0064	0.0053	0.0049	0.0072	0.0048	0.0034
ln SIZE					0.0060*	0.0076*	0.0099*	0.0097*
PROD					−0.0128*	−0.0093*	−0.0072*	−0.0023
PROF					−0.0012	−0.0015	−0.0040	−0.0068*
GROWTH					0.0062*	0.0009	−0.0022	−0.0045
CONSTANT	−0.4678*	−0.4684*	−0.4686*	−0.4729*	−0.4706*	−0.4703*	−0.4714*	−0.4761*
Panel B: model performance								
Brier score	0.0333	0.0336	0.0336	0.0332	0.0330	0.0334	0.0335	0.0330
Threshold	0.0313	0.0321	0.0320	0.0300	0.0333	0.0323	0.0324	0.0353
Type I error	30.3500	31.9300	23.1650	18.2950	36.5700	29.0950	27.9700	25.3400
Type II error	1,541.61	1,425.8000	1,454.2050	1,137.6650	1,322.4950	1,466.0850	1,425.7400	886.7400
% Correct default	0.7629	0.7635	0.8333	0.8603	0.7143	0.7845	0.7988	0.8066
% Correct non default	0.5712	0.6209	0.6250	0.6880	0.6321	0.6102	0.6324	0.7568
Panel C: model performance								
Threshold					0.0313	0.0321	0.0320	0.0300
Type I error					29.1150	28.5450	26.7150	17.4650
Type II error					1,542.1300	1,483.1800	1,459.9450	1,152.6000
% Correct default					0.7725	0.7886	0.8078	0.8667
% Correct non default					0.5710	0.6056	0.6235	0.6839

Variables are in z-scores. *Panel A*: Bootstrap means of marginal effects at the sample average of covariates. *Significant at 1% level. *Panels B and C*: Bootstrap means of model performance measures

The findings in the right part of the table, obtained from specification in Eq. 5, confirm the predominant role played by cost of debt among the financial indicators, but also offer strong support to the idea that economic characteristics of firms have a relevant effect, additional to that of financial variables. When significant, the signs of economic indicators are consistent with the foregoing evidence on kernel densities and stochastic dominance, discussed in Sections 3 and 4. Size and Growth have indeed a positive effect, while Productivity and Profitability reduce the probability of default. Also notice intertemporal variation of the effects, revealed by varying magnitude of estimates, together with patterns of significance. In particular, Size is strongly significant in all years and the effect seems increasing over time. The marginal effect of Productivity is instead decreasing over time, and looses its significance in the last year. There might be an interaction with Profitability, which indeed turns significant in 2002, and with a relatively big negative coefficient, possibly "absorbing" part of the loosing significance of Productivity in this same year. The role of Growth seems instead marginal, with a moderate effect significant in the first year only.

Overall, we can conclude that statistical relevance of economic variables is preserved even at very short time distance to default. Both financial and economic dimensions of firm operation, in other words, matter over the longer as well as over the shorter run. This suggests existence of strong capital market imperfections, preventing to consider financial structure of firms as embedding all the relevant information on the probability that a firm incur default. Rather, industrial characteristics and performances capture important and complementary determinants of default, even in the short run. The bootstrap procedure, with its random sampling of firms, allows to safely conclude that the observed intertemporal variation cannot be simply attributed to outliers or missing observations affecting the estimates in each specific year.[20]

A further test of the contribution offered by economic variables is conducted in panel B of the same Table 3, where we focus on goodness of fit and prediction accuracy of the models.

The first measure adopted is the Brier Score (Brier 1950), a standard indicator employed in the literature to assess the relative explanatory power of alternative models of distress prediction. For each firm i, this is computed as $1/N \sum_{i=1}^{N} (Y_i - P_i)^2$, where N is the number of firms, P_i is the estimated probability of default of firm i based on coefficient estimates of the probit regressions, and Y_i is the actual realization of Y for firm i, default ($Y_i = 1$) or non-default ($Y_i = 0$). We report the bootstrap mean of this measure over the 200 replications. Of course, the lower the Brier score, and the higher the performance of the model. Thus, comparisons of results between corresponding specifications—i.e. looking at each "financial variables only" model vs. the "financial plus economic variables" model estimated in the same year—

[20] Also notice that the main results persist if we perform bootstrap estimate of linear probability or logit models, and do not change if we take the number of employees as a proxy for size.

strongly confirm that inclusion of economic variables provides an improved model performance in all years.

Prediction accuracy of the models is then evaluated building upon the concept of correctly classified observations. This is based on the idea of classifying a firm as defaulted (assigning $Y = 1$) whenever its estimated probability of default, P_i, lies above than a certain threshold value τ, while a firm is classified as non-defaulting otherwise (assigning $Y = 0$). Such a classification will in general differ from the true default or non-default status. A Type I error is defined as the case when a firm which is actually defaulting (a true $Y_i = 1$ in the data) is classified as non-defaulting, and thus is assigned a 0. A Type II error is instead counted when a non-defaulting firm (a true $Y_i = 0$) is assigned a 1 by the classification procedure. Correspondingly, the percentage of correctly predicted 1's ("% Correct default" in the table) gives the ratio of the correctly predicted defaults over the actual number of defaults in the sample. Conversely, the percentage of correctly predicted 0's ("% Correct non default" in the table) gives the fraction of correctly classified non-defaulting firms over the actual number of non-defaulters. Within this set of measures, it is standard to prefer models reducing Type I errors (or maximizing the percentage of correctly predicted defaults). Indeed, from the point of view of an investor, failing to predict a bankruptcy (and investing) might be much more costly than mistakenly predicting a default (and not investing).

Quite obviously, the degree of prediction accuracy depends on the specific value of the threshold τ. Different criteria are in principle available to set this value. We consider an "optimal" τ^* so as to minimize the overall number of prediction errors (Type I plus Type II), weighted by the relative frequency of zeros and ones. This is obtained in practice by solving the following minimization problem

$$\tau^* = \arg\min_{\tau} \left(\frac{1}{N_0} \sum_{i \in ND} \Theta(P_i - \tau) + \frac{1}{N_1} \sum_{i \in D} \Theta(\tau - P_i) \right) \quad , \tag{7}$$

where $\Theta(x)$ is the Heaviside step function, taking value $\Theta(x) = 0$ if $x < 0$ and $\Theta(x) = 1$ if $x > 0$, while N_0 and N_1 stand for the actual number of non-defaulting and defaulting firms in our sample, respectively, with ND and D the two corresponding sets.[21] We repeat the minimization procedure for each bootstrap replication, and then compute Types I and II errors, together with the percentage of correctly predicted default and non default events.

In panel B we report averages of these measures computed over the 200 bootstrap replications, together with the average value of the optimal

[21] Minimizing the overall number of errors is equivalent to maximize the total sum of correctly predicted observations. The weighting is instead introduced to address the specific characteristics of our exercise. True 0's are indeed much more frequent than true 1's, simply because default rates in each bootstrapped sample equal the population-wide frequencies presented in Table 1.

threshold found at each run.[22] Results suggests that the models including economic variables tend to produce more Type I errors (and thus lower number of Type II errors) as compared to corresponding "financial variables only" specifications. It is however important to underline that a direct comparison of these numbers is not truly informative, as they are obtained with different values of τ^*, each optimal for its own model. A more meaningful comparison between two models requires to evaluate the performance of one model under the optimal threshold of the other model. This is done in panel C of Table 3. Here prediction accuracy of each "financial plus economic variables" specifications is computed taking the optimal threshold τ^* of the "financial variables only" model estimated in the same year: if the former performs better under the τ^* of the latter, this would imply a strong confirmation that including economic variables into the analysis is improving predictive power with respect to the benchmark "financial variables only" specification. What we observe is that, compared to the benchmark figures of panel B, the models including economic variables perform better in terms of Type I errors (and correctly classified defaults), in all the years but 2001. This provides further evidence that, consistently with suggestions derived from Brier Scores, the contribution of economic variables is important. Once again, the improved performance in 2002 confirms that this holds true even in the very proximity of the default event.

6 Distance to default and credit ratings

Following the literature on corporate default prediction, there are two further measures which one should consider in the analysis of default probability, Distance to Default (DD) and credit ratings.

Despite theoretically appealing, DD has two major limitations. First, due to the non trivial procedures required to get a numerical solution of the models, computation of this measure is in practice rather complicated. Second, the measure applies to publicly traded firms only, because the underlying market value of a firm, not observable in practice, is approximated through market value of equity, essentially exploiting the standard hypothesis that markets are fully informed and stock prices instantaneously incorporate all the relevant information. A solution to the first problem is to adopt a "naive" DD measure (Bharath and Shumway 2008), which is much easier to compute than the original DD and ensures, at the same time, equivalent results in terms of default prediction accuracy. Yet, the naive DD still requires data on market values of firms' equity and assets, so that it is not obvious how it is possible to include this measure in the context of our study, where the

[22]The application of the bootstrap to compute model performance measures is particularly important. Zmijevski (1984) indeed shows that classification and prediction errors of the defaulting group are generally overstated without an appropriate treatment of the "choice-based sample" problem.

scope of analysis goes beyond the limited subset of publicly traded firms, and, consequently, only accounting data are available. Nevertheless, motivated by the widespread use and the solid theoretical basis of DD, we attempt to include such a potentially important explanatory variable in the analysis. We start from the naive DD estimator of Bharath and Shumway (2008) and build an equivalent measure, labeled as Book Distance to Default (BookDD), based on accounting data on value of shares and value of debt, which we can derive from available figures on Leverage and Total Assets. Notice that computations exploit time series means and volatility of these variables, and thus BookDD takes a single value for each firm, not varying over time (see the Appendix for details on construction of the proxy).

We explore estimates of a first model where BookDD enters as the sole covariate

$$P(Y_T = 1 \mid X_t) = \Phi(\beta_0 + \delta\ BookDD), \tag{8}$$

and then add the full set of financial and economic variables considered in this work

$$\begin{aligned} P(Y_T = 1 \mid X_t) = \Phi \Bigg(& \beta_{0t} + \delta\ BookDD + \beta_{1t}\ \frac{IE_t}{S_t} + \beta_{2t}\ LEV_t \\ & + \beta_{3t}\ \frac{FD_t}{S_t} + \beta_{4t}\ S_t + \beta_{5t}\ PROD_t \\ & + \beta_{6t}\ PROF_t + \beta_{7t}\ GROWTH_t \Bigg). \end{aligned} \tag{9}$$

The estimation strategy goes exactly as in previous section.[23]

Results, in Table 4, are clearcut. We find that BookDD has a tight link with default: marginal effects are big and always statistically significant. However, the inclusion of this further regressor does not affect any of the results achieved in the foregoing section: the sign, magnitude and temporal patterns of statistical significance of the effects of financial and economic variables remain in practice unchanged as compared to our baseline results in Table 3. We only observe that BookDD absorbs Leverage and FD/S in some years, but this is not surprising as equity and debts enter the definition of BookDD. Moreover, the goodness of fit measures in panel B show that the inclusion of BookDD does not produce any substantial improvement in the predictive ability of the model: Brier score and both types of error are indeed comparable with the ones reported in Table 3. In the same direction, panel C shows that the inclusion of economic and financial variables yield predictions which are noticeably better than those obtained using BookDD alone.

[23] Statistical irrelevance of sectoral dynamics motivate the exclusion of 2-Digit dummies from the exercise.

Table 4 Probit estimates of default probabilities by year, including Book DD as modeled in Eqs. 8 and 9—results over 200 bootstrap replications

	Bootstrap probit with distance to default—estimates by year							
	BookDD only model				BookDD, financial and economic variables			
	(1) 1999	(2) 2000	(3) 2001	(4) 2002	(5) 1999	(6) 2000	(7) 2001	(8) 2002
Panel A: estimates								
IE/S					0.0074*	0.0066*	0.0086*	0.0090*
LEV					0.0005	0.0015	0.0002	0.0029
FD/S					0.0024	0.0053	0.0021	0.0026
ln SIZE					0.0066*	0.0076*	0.0075*	0.0072*
PROD					−0.0097*	−0.0067*	−0.0041	−0.0010
PROF					0.0013	−0.0005	−0.0024	−0.0049*
GROWTH					0.0053*	0.0004	−0.0019	−0.0020
Book DD	−0.0210*	−0.0204*	−0.0898*	−0.0816*	−0.0164*	−0.0147*	−0.1095*	−0.1064*
CONSTANT	−0.4698*	−0.4693*	−0.4764*	−0.4766*	−0.4733*	−0.4726*	−0.4800*	−0.4811*
Panel B: model performance								
Brier score	0.0332	0.0335	0.0335	0.0335	0.0329	0.0334	0.0333	0.0331
Threshold	0.0374	0.0385	0.0364	0.0362	0.0343	0.0315	0.0345	0.0361
Type I error	43.9900	51.6650	52.2200	49.2350	32.6450	22.6450	30.2400	29.2500
Type II error	1,224.4550	1,180.9650	1,305.3200	1,358.4750	1,159.7000	1,431.3350	1,131.6050	901.6500
% Correct default	0.6364	0.5964	0.6014	0.6242	0.7302	0.8231	0.7692	0.7767
% Correct non default	0.6386	0.6677	0.6420	0.6274	0.6577	0.5973	0.6896	0.7527
Panel C: comparisons of prediction performance against the "rating only" model of the same year								
Threshold					0.0374	0.0385	0.0364	0.0362
Type I error					41.6750	46.8650	34.7800	29.5750
Type II error					951.9550	974.0700	1,020.3050	892.4400
% Correct default					0.6556	0.6339	0.7345	0.7742
% Correct non default					0.7190	0.7259	0.7202	0.7552

Variables are in *z*-scores. *Panel A*: Bootstrap means of marginal effects at the sample average of covariates. *Significant at 1% level. *Panels B and C*: Bootstrap means of model performance measures

The second extension of the model we want to consider in this section includes the CeBi credit rating among the regressors. Credit ratings, in general, represent a short-run forecast of default probability, hence their inclusion could help to validate the statistical consistence of the timing effects discovered so far. However, a first advantage of the CeBi ratings over credit files from international rating agencies lies in that the CeBi rating is available for all the firms included in the dataset. On the contrary, international agencies are mainly concerned with bigger Italian firms, those having reached an international relevance, and/or listed on stock exchanges around the world, a fact that would bias the scope of analysis towards a non representative set of firms. Another peculiar characteristic of the CeBi index is that it is an official rating. Indeed, CeBi has a long-standing tradition as an institutional player within the Italian financial system. Founded as an agency of the Bank of Italy in the early 80's, credit rating construction has been since then one of the core activities of CeBi within the institutional tasks of providing assistance in banking system supervision. Today a private company, CeBi is still carrying out an institutional mandate, as the Italian member within the European Committee of Central Balance Sheet Data Offices (ECCBSO), operating in close relationships with the Italian Statistical Office and the major commercial banks. These observations allow us to be confident that the CeBi index is reliable and maintained up to international standards.

The firms included in the database, no matter whether defaulting or non-defaulting at the end of the period, are ranked with a score ranging from 1 to 9, in increasing order of financial fragility: 1 is attributed to highly solvable firms, while 9 identifies firms displaying a serious risk of default, and this ranking is an ordinal one.

For our purposes, we group firms into three classes, which we label Low Rate firms (having lower probability of default, with credit ratings 1–6), Mid Rate firms (rated 7) and High Rate firms (rated 8–9). The transition matrices among the three groups, displayed in Table 5, summarize the salient properties of the rating index in the sample. Over the longer-run transition (from 1998 to the default year), the Low Rate class is very stable, while both Mid and High Rate firms display a sort of "reversion to the mean" property, i.e. they have a higher probability to jump back to better ratings, as compared to the likelihood of remaining in the same class. This gets reflected in the transition probability to end up defaulting (last column), which is higher for Mid Rate

Table 5 Credit ratings transition matrices

		Default year			
		Low	Mid	High	Default
1998	Low	0.8948	0.0824	0.0176	0.0052
	Mid	0.5402	0.3651	0.0743	0.0204
	High	0.4589	0.3290	0.1991	0.0130
2002	Low	0.9308	0.0583	0.0077	0.0032
	Mid	0.3375	0.5446	0.0986	0.0192
	High	0.1499	0.3512	0.4754	0.0236

firms than for High Rate firms. Similar patterns persist in the short run transition (from 2002 to default year). The numbers on the diagonal suggest higher stability within-class, as compared to the longer-run transition. Yet, the "reversion to the mean" property—towards improved ratings—is still present, even in the High Rate group. Notice however that the short-run transition probabilities to default (last column) are more in accordance with what one would expect: probability of default increases as rating worsens. This confirms the presumption that credit ratings are much better predictors of default in the very short run than over a longer distance to the event. In turn, the fact that the CeBi index displays variation over time is important, allowing to test the time effects of financial and economic variables observed in the foregoing probit regressions.[24]

According to the classification in three rating classes, we build three dummy variables taking on value 1 when a firm is belonging to one of the classes, and zero otherwise. These are then employed in order to investigate if inclusion of different credit rating conditions is able to affect the conclusions drawn from the baseline year by year estimates presented in the previous section. That is, running separate regressions for each year over the 1999-2002 sample period, we first estimate a "rating only" specification

$$\text{Prob}\,(Y_T = 1 \mid X_t) = \Phi(\beta_{0t} + \delta_{1t}\ LOW_t + \delta_{2t}\ MID_t + \delta_{3t}\ HIGH_t), \quad (10)$$

allowing to get an idea of the explanatory power of the CeBi index, and then compare results with a second model where financial and economic characteristics enter together with ratings themselves

$$\begin{aligned}\text{Prob}\,(Y_T = 1 \mid X) = \Phi\Bigg(&\beta_{0t} + \delta_{1t}\ LOW_t + \delta_{2t}\ MID_t + \delta_{3t}\ HIGH_t + \beta_{1t}\ \frac{IE_t}{S_t} \\ &+ \beta_{2t}\ LEV_t + \beta_{3t}\ \frac{FD_t}{S_t} + \beta_{4t}\ S_t\beta_{5t}\ PROD_t \\ &+ \beta_{6t}\ PROF_t + \beta_{7t}\ GROWTH_t\Bigg). \end{aligned} \quad (11)$$

Table 6 shows the results. Due to obvious collinearity between the rating dummies and the constant term we present coefficient estimates of regressions where only the Low and Mid Rate class are considered.[25]

[24]Notice that firms' "ability" to improve their rating does not depend on the exit of better firms from the sample. The matrices are indeed computed taking all the firms which are still in the sample in the last year, when default is measured, and then tracing back their credit rating history. The findings reported in Bottazzi et al. (2008) show that a similar intertemporal behavior in the CeBi index is also appearing when a different division of firms into rating classes is chosen.

[25]Once again, statistical irrelevance of sectoral dynamics motivate the exclusion of 2-Digit dummies from the models.

Table 6 Probit estimates of default probabilities including credit ratings as modeled in Eqs. 10 and 11—results over 200 bootstrap replications

	Bootstrap probit with credit ratings—estimates by year							
	Rating only				Rating, financial and economic			
	(1) 1999	(2) 2000	(3) 2001	(4) 2002	(5) 1999	(6) 2000	(7) 2001	(8) 2002
Panel A: estimates								
IE/S					0.0039*	0.0041*	0.0049	0.0088*
LEV					−0.0001	−0.0013	−0.0015	0.0028*
FD/S					0.0044	0.0059	0.0026	0.0019
ln SIZE					0.0061*	0.0072*	0.0090*	0.0086*
PROD					−0.0111*	−0.0075*	−0.0059*	−0.0011
PROF					−0.0004	−0.0015	−0.0021	−0.0037*
GROWTH					0.0054*	−0.0000	−0.0038*	−0.0036*
CONSTANT	−0.3694*	−0.4770*	−0.3720*	−0.1628*	−0.4999*	−0.6582*	−0.4960*	−0.3021*
LOW	−0.0301*	−0.0076	−0.0323*	−0.1206*	0.0001	0.0186*	−0.0047	−0.0428*
MID	0.0198*	0.0614*	0.0414*	−0.0072	0.0392*	0.1134*	0.0639*	0.0068
Panel B: model performance								
Brier score	0.0328	0.0327	0.0322	0.0319	0.0325	0.0326	0.0320	0.0316
Threshold	0.0258	0.0284	0.0193	0.0163	0.0290	0.0254	0.0239	0.0270
Type I error	78.0000	70.7850	61.0700	48.7550	36.6350	30.0450	29.8300	25.8250
Type II error	576.1900	616.3500	709.3900	633.7250	1,248.9350	1,334.5550	1,230.2100	901.1550
% Correct default	0.3906	0.4757	0.5606	0.6278	0.7138	0.7774	0.7854	0.8029
% Correct non default	0.8397	0.8361	0.8171	0.8262	0.6526	0.6452	0.6828	0.7528
Panel C: comparisons of prediction performance against the “rating only” model of the same year								
Threshold					0.0258	0.0284	0.0193	0.0163
Type I error					25.2550	39.5450	16.2850	11.8150
Type II error					1,612.8600	1,111.9150	1,722.5850	1,507.6900
% Correct default					0.8027	0.7071	0.8828	0.9098
% Correct non default					0.5514	0.7044	0.5558	0.5865

Variables are in z-scores. *Panel A*: Bootstrap means of marginal effects at the sample average of covariates. *Significant at 1% level. *Panels B and C*: Bootstrap means of model performance measures

Columns 1–4 present the estimates of regressions where credit ratings are considered alone. The signs and magnitudes of constant term and dummies are consistent with the intertemporal variation of CeBi ratings suggested by transition probability matrices. Low and Mid dummy coefficients essentially depends on the relative proportion of defaulters in these classes, as compared to the proportion of defaulters in the High Rate class (this latter influencing the value of the constant term). The negative estimates of the Low Rate dummy reflects the lower percentage of defaults among Low Rate firms, while positive Mid Rate dummy reflects the "reversion to the mean" effect discussed above. Also notice that model performance measures (cfr. panel B) tend to confirm the 1-year ahead forecast nature of the index: Brier Scores and Type I errors improve approaching the default time, and the model of 2002 is the one achieving the best performance records.

The models where we add the other covariates (cfr. columns 5–8) display very similar patterns concerning the estimated effects of rating dummies, corroborating the corresponding "rating only" models. More importantly, the effects of financial and economic variables are broadly surviving the inclusion of credit ratings. Credit ratings are surely playing a role, as indeed marginal effects of financial and economic variables are lower as compared to the benchmark estimates without credit dummies (cfr. once again Table 3). Still, we are able to confirm the main conclusions drawn in the previous section. First, the cost of debt is still playing the main role among financial characteristics, also displaying the expected positive effect, increasing over time. Second, economic factors tend to retain their signs. The positive and strongly significant impact of size is confirmed, as well as the negative effect of productivity and profitability, with the former significant in first three years, and the latter significant in 2002, exactly as before. The negative and significant role of growth in the last years differs from previous estimates without ratings. This is probably related to instabilities in the tail behavior of this variable across defaulters and non-defaulters, noticed when commenting non parametric distributional properties. Third, and lastly, the findings here reproduce the same time effects observed before. Economic variables, as a whole, are thus confirmed to play an important role as default predictors, in addition to financial conditions, over both the longer and shorter run. The main effect induced by including credit ratings seems to be on model performance (cfr. panel B). As compared to models where credit ratings were not included (compare again with Table 3) we observe lower Brier Scores. These improvements, however, are not entirely accounted for by ratings alone. Indeed, models with the full set of regressors (rate dummies plus economic and financial variables) tend to perform better than the corresponding "rating only" models: when we compute prediction errors taking the optimal τ^* of the corresponding "rating only" models in the same year (see panel C), Brier Scores are lower and Type I errors decrease considerably.

Summarizing, the analyses of this section validate the robustness of our findings. Both Distance to Default and credit ratings can surely complement for some of the relevant dimensions which were not considered in the previous

section. Nevertheless, the effects exerted by the set of financial and economic characteristics remain valid, and we can confirm the complementary and additional statistical relevance of economic variables over both the longer and shorter time horizons allowed by the dataset.

7 Conclusion

This paper provides an empirical investigation of the determinants of firm default. Default, formally defined as firm's inability to repay commercial and financial debts, has been the object of extensive analysis within financial economics. The relation of default with bankruptcy and other forms of involuntary exit makes it a reliable alternative to business registers' events for the identification of business failures. While the latter are indeed often affected by difficult interpretation and approximate timing, default events represent clearly defined signals of firm distress and are precisely identified in time.

It is commonly accepted in the literature that financial factors should be able, at least to a major extent, to capture the main determinants of default. And this should hold especially when the time of default is approaching, given the conjecture that present financial variables reflect all the past information on the industrial or economic performances of firms. Our findings show that, on the contrary, economic characteristics of firms do play a relevant role, both over the longer and the shorter run.

First, distributional properties show that will-be defaulting firms differ from non-defaulting firms not only on the grounds of their financial condition, but also with respect to their industrial performances, at different time distances to default. Defaulting firms are on average more financially exposed, but they are also less productive and less profitable in all the years before the default occurs. Interestingly, size and growth display a positive relationship with default, although distributional differences are less marked and mainly related with tail behavior. Tests of distributional equality confirm the positive association of size with probability of default, while the role of growth appears negligible.

Second, probit models reveal that the two sets of variables, financial and economic, capture diverse, albeit complementary, determinants of the process leading to firms' distress. Cost of debt exerts the most important effect among the financial variables. At the same time, higher levels of productivity and profitability reduce the probability of default, with the latter more relevant when firms are very close to default and the former more significant in the previous years. In agreement with the non-parametric analysis, size displays a relatively big, persistent over time, impact on default occurrences, increasing their likelihood. The effect of growth is positive but barely significant.

Our findings are robust under several respects. First, the bootstrap procedure employed warrants us that estimates are not affected by under-weighting of defaulting firms in the sample as compared to the actual default rates observed at the national level. Second, the main patterns remain valid when

we include, in the probit equations, a theoretically based measure of Distance to Default and a credit rating index. The latter accounts, by construction, for dimensions of firm dynamics which we might not completely capture with our set of main regressors. Finally, we find that, although firms acting in different sectors are expected to be highly heterogeneous with respect to both financial and economic characteristics, sectoral specificities (at the 2-Digit level) do not seem to affect the link between such characteristics and the default probability.

Our findings contribute to both industrial and financial economics. The negative effect of productivity and profitability on default probability is in agreement with the inverse relationship of these variables with exit usually observed in the industrial literature. Conversely, the positive effect of size contrasts with this literature, as one typically finds that the probability of exit decreases with size. Though less intuitive, the result is strong, emerging from all the different statistical analysis proposed in this work. One could suspect that the positive effect of size is simply due to the fact that our defaulters, belonging to the group of firms having a formal credit relationship with a large commercial bank, are bigger than the average firm in the sample. However, we do not observe over-representation of small firms in the non-defaulting group. Differences in the size effect between our analysis and the majority of studies in industrial economics probably rest in the adoption of different definitions of business failure. Our results show that size does not prevent from incurring into solvency problems, even as severe as default. However, due to factors like reputation or long-lasting relationships with financial and public institutions, big firms can be less likely to transform a temporary financial distress into radical restructuring or to initiate a formal bankruptcy procedure, whence the reduced likelihood to finally incur into an (involuntary) exit event (see Bottazzi and Tamagni 2011).

With reference to the financial literature, this paper attempts to yield empirical validation to the intuition, so far seldomly tested, that default cannot be regarded as a mere financial phenomenon. Economic variables indeed exhibit strong statistical significance in the years before default occurs, and their contribution to the predictive power of models does not vanish when we consider the short run. This finding suggests the existence of severe capital markets imperfections which create a wedge between financial and industrial characteristics of the firm, even in the short run, when financial variables are commonly conceived to be strong predictors of default. The result, beyond contrasting with standard theory assuming complete and fully informed markets, has relevant policy implications. One can indeed conclude that the accuracy of standard risk assessment devices, such as official credit ratings or risk management procedures internally maintained by financial institutions, might be biased, devoting too little attention to important industrial/economic factors. Our results, hence, support the need to develop broader, multidimensional approaches to the evaluation of corporate default risk, enlarging the scope of the dimensions commonly included, and placing specific attention to the different time horizons considered.

Appendix: Costruction of book distance to default

As explained in the text, we start from the *naive DD* of Bharath and Shumway (2008). For each firm, this is defined as

$$\text{naive DD} = \frac{\ln[(E+F)/F] + \left(r_{i,t-1} - 0.5\ \text{naive}\ \sigma_V^2\right) T}{\text{naive}\ \sigma_V \sqrt{T}}, \tag{12}$$

where E is market value of equity, F is face value of debt, T is the time-to-maturity assuming each firm has issued just one bond maturing in T periods, $r_{i,t-1}$ is the firm's stock return over the previous year, and naive σ_V is the volatility of the value of the firms, computed as

$$\text{naive}\ \sigma_V = \frac{E}{E+F}\sigma_E + \frac{F}{E+F}(0.05 + 0.25\sigma_E), \tag{13}$$

with the last term in parenthesis being an estimate of the volatility of firm debt

$$\text{naive}\ \sigma_F = 0.05 + 0.25\sigma_E. \tag{14}$$

This default predictor involves a computationally easier estimate of the underlying value of a firm as compared to numerical solution of Black-Scholes-Merton's equations, and its predictive power of default has been found to be comparable to that of the original DD measure.

To make this definition operational in the context of our dataset, where most of the firms are not publicly traded, we make the following choices, based on available accounting book variables. First, we place the time of computation in 2002, the last year before default is measured in our data, so that T = 1. Second, E is proxied with the sum of annual income after taxes plus face value of outstanding shares, which we define Book Equity, BE. This is simply the denominator of our measure of Leverage, and therefore we can compute it by

$$BE = \text{Total Assets/Leverage}. \tag{15}$$

Third, since Total Assets equals the sum of BE plus the stock of outstanding debt, due to Italian accounting practices, we can proxy F via

$$D = \text{Total Assets} - BE. \tag{16}$$

Fourth, in place of $r_{i,t-1}$, we take the time series average of firm i's growth rates of Book Equity, μ_{BE}, computed over the years before default occurs (1999–2002). This smoothing is done to incorporate all available past information, which is what the *naive DD* assumes to be entirely captured by stock returns over the previous year, due to efficient and fully informed stock markets. Fifth, in place of the approximation of debt volatility contained in Eq. 14, we directly compute the volatility of D, as the standard deviation of the growth rates of D over the years 1999–2002. Finally, the same is done for σ_{BE},

the volatility of Book Equity. Therefore, our "accounting book version" of the naive DD becomes

$$\text{BookDD} = \frac{\ln[(BE_{2002} + D_{2002})/D_{2002}] + \left(\mu_{BE} - 0.5 \text{ naive } \sigma_V^2\right)}{\text{naive } \sigma_V}, \tag{17}$$

with

$$\text{naive } \sigma_V = \frac{BE}{BE + D}\sigma_{BE} + \frac{D}{BE + D}\sigma_D. \tag{18}$$

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